Winfree

phase only

$$d\theta_i/dt = \omega_i + Z(\theta_i)\ \Sigma_j\ P(\theta_j)$$

influence P and sensitivity Z are separate functions

WONN

Kuramoto

phase only

$$d\theta_i/dt = \omega_i + (K/N)\ \Sigma_j \sin(\theta_j - \theta_i)$$

a single sinusoid of the phase difference

Un-0 · KomplexNet · Kuramoto Attention · Kuramoto Orientation Diffusion

Kuramoto–Sakaguchi

phase only

$$d\theta_i/dt = \omega_i + (K/N)\ \Sigma_j \sin(\theta_j - \theta_i - \alpha)$$

the same sinusoid, carrying a phase lag α

FSN

Daido harmonics, with delay

phase only

$$d\theta_i/dt = \omega_i + \Sigma_j\ \Gamma(\theta_j(t - \tau) - \theta_i)$$

an arbitrary periodic Γ, and coupling that depends on history

FSN

D-dimensional Kuramoto

phase on a sphere

$$dx_i/dt = \Omega_i x_i + P_{x_i} (c_i + \Sigma_j J_{ij} x_j)$$

unit vectors in D dimensions; the phase leaves the circle

AKOrN

Stuart–Landau

amplitude and phase

$$dz_i/dt = (\mu + i\omega_i)z_i - |z_i|^2 z_i + K\ \Sigma_j (z_j - z_i)$$

complex state: amplitude and phase both evolve

Zhang et al. 2026

Damped second-order

no coupling term

$$y'' = \sigma(Wy + W'y' + Vu + b) - \gamma y - \varepsilon y'$$

an oscillatory recurrence; damping bounds the gradients

coRNN · UnICORNN · RON · Neural Wave Machines

Linear oscillatory state space

no coupling term

$$y'' = -Ay - Gy' + Bu,\ A,\ G \text{ diagonal}$$

linear, so the whole sequence solves by parallel scan

LinOSS · D-LinOSS

Physical substrates run their own device dynamics and are written as none of these forms: the spintronic reservoir, and oscillator Ising machines.